PROJECT REPORT

Smart Farm Guard:: Protectting Crotps from Wildlife.

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ABSTRACT

Crop damage caused by wild animals is a major problem faced by farmers, especially in rural and forest-bordering areas. This project aims to develop an automated system that detects wildlife approaching farmland and activates deterrent methods to protect crops. The system uses sensors such as PIR or ultrasonic sensors to monitor movement around the field. When an animal is detected, the microcontroller processes the signal and triggers alarms, lights, or other harmless deterrents to scare the animal away. The design focuses on providing a low-cost, energy-efficient, and reliable solution suitable for small and medium farmers. The system helps reduce human effort, increases safety during night hours, and minimizes crop loss. Overall, the project demonstrates a simple yet effective method to support farmers through technology-based crop protection.

KEYWORDS/ ABBREVIATION:

PIR – Passive Infrared Sensor

LED – Light Emitting Diode

IoT – Internet of Things

*Wildlife Detection

* Crop Protection System

* PIR Sensor

CHAPTER 1

INTRODUCTION

- Agriculture is important for every country, and farmers work very hard to grow crops. But a major problem they face today is the damage caused by wild animals. Animals like pigs, monkeys, deer, cows, and other wildlife enter the fields and destroy crops.

- This leads to big losses for farmers and affects their income. Guarding the field all the time is very difficult, especially at night. Traditional methods like fences, fire, or shouting do not always work.
- To solve this problem, technology can be used in a simple and affordable way. Sensors and small electronic devices can help detect animals when they come near the crops. An alarm or buzzer can alert the farmer so the animals can be stopped before they cause damage.
- This project," Crop Protection from Wildlife", aims to create an easy and low-cost system that helps farmers protect their crops using sensors and an automatic alert mechanism.
- The system works by detecting movement using PIR or ultrasonic sensors. When an animal is detected, the microcontroller activates a buzzer or light. This helps scare the animal away and warns the farmer. The system is safe, does not harm animals, and reduces the workload of farmers. It is also useful in remote areas where constant monitoring is difficult. This project shows how simple technology can solve real-life farming problems**.**

Working Principle

1. Detection: Sensors placed around the field detect the movement of animals when they come close to the crops.

2.*Activation:* Once an animal is detected, the system automatically turns on alarms, lights, or buzzers.

3. *Scaring the Animal Away:* The sudden sound or light scares the animal, making it run away from the field.

4. *Alert to Farmer :* The system can also send an alert to the farmer's phone if connected.

5.*Continuous Monitoring:* It works day and night, regularly monitoring the field without any manual effort

Advantages

- Reduce crop damage by keeping wildlife away

- Save time and effort with automatic monitoring
- Cost effective
- Work continuously even at night

Project Aim and Objective:

•**Aim**:

The main aim of this project is to design and develop a simple, effective, and low - cost system that protects crops from wildlife animals using sensors and automatic alert mechanism

•**Objectives**:

- To detect the presence of animals near agricultural fields using sensors.
- To alert the farmer immediately through buzzer when an animal is detected.
- To reduce crop loss by preventing animals from entering the farmland.
- To develop a system that affordable, easy to install, and requires low maintenance.
- To create awareness about use of technology in modern farming.
- To minimize human wildlife conflicts by using safe and non-harmful methods.

•Problem Statement:

Farmers face huge losses because wild animals enter farms and damage crops. Manual monitoring is difficult, especially at night. So, there is a need for simple and low-cost system that can detect animals and protect crops.

• Problem Description*:*

Farmers face major losses every year because wild animals such as monkeys, wild boars, deer, and birds enter the crop fields and damage the plants. These animals destroy growing crops, eat the yield, and sometimes even harm farmers. Traditional methods like fencing, shouting, firecrackers, or scarecrows are not always effective. They require continuous human effort and cannot protect the field during night hours or when the farmer is away.

Due to this, farmers suffer reduced crop production, financial loss, and increased stress. There is a need for a *simple, affordable, and reliable system* that can automatically detect the presence of wildlife near the field and protect the crops without harming the animals. The system should alert the farmer in time or use safe methods like sound, lights, or sensors to scare animals away.

Therefore, the project aims to design a *smart crop protection system* that helps farmers safeguard their fields continuously, reduces manual work, prevents crop damage, and ensures better productivity.

• **Software Requirements**:

Arduino IDE

Hardware Requirements:

7.1 ESP32

* Acts as the brain of the system, controlling sensors and output devices.
* Processes signals from the sensors and activates alarms/lights accordingly.
* Easy to program and widely used for small automation projects.

7.2 PIR Motion Sensor

* Detects movement of animals in the field using infrared radiation.
* Sends a HIGH signal to the microcontroller when motion is detected.
* Works continuously and is energy-efficient.

7.3 Ultrasonic Sensor (Optional)

* Measures distance to nearby objects or animals using ultrasonic waves.
* Can detect animals before they reach the crops for early warning.
* Provides more accurate detection in open fields compared to PIR alone.

7.4 Buzzer

* Produces loud sound to scare animals away from the crops.
* Controlled by the microcontroller based on sensor input.
* Low-cost and effective for short-range deterrence.

7.5 LED / Light Indicator

* Flashes or glows to alert nearby wildlife and scare them.
* Can also act as a visual alert for the farmer.
* Simple to connect and uses very low power.

7.6 Power Supply (5V DC Adapter or Battery)

* Provides stable power to the microcontroller and sensors.
* Can be a battery for portability or a DC adapter for continuous use.
* Ensures the system works day and night without interruption.

7.7 Connecting Wires / Breadboard

* Used to connect all components to the microcontroller without soldering.
* Makes circuit adjustments easy during testing and prototyping.
* Allows neat and organized circuit design.

7.8 Resistors (if needed)

* Control current flowing to LEDs or sensors to prevent damage.
* Ensure the system works safely and reliably.

CHAPTER 2

LITERATURE SURVEY

1. Introduction

Crop protection from wildlife has become an important area of research in modern agriculture. Farmers face huge losses every year because of animals

such as wild boars, deer, monkeys, elephants, and birds entering farms and damaging standing crops. Traditional methods like manual guarding, scarecrows, and fencing are not always effective, especially during night hours. Therefore, researchers and engineers have started developing *automated, sensor-based and technology-driven systems* to protect farmland in a smarter and more reliable way.

Over the past few years, literature has shown a clear shift toward using *electronics, embedded systems, artificial intelligence, and IoT* to monitor fields and prevent wildlife intrusion. Most studies focus on designing systems that can *detect animals early, **alert the farmer, and **use harmless deterrent techniques* to scare the animals away without causing injury.

2. Sensor-Based Detection Systems

Many research papers highlight the use of *sensors* as the heart of wildlife protection systems.

The most common sensors found in literature are:

• PIR (Passive Infrared) Sensors*

These sensors detect body heat and movement of animals. They are widely used because they are cheap, easy to install, and work well for night-time detection. Many prototypes combine PIR sensors with buzzers or lights to create a quick alarm system.

• Ultrasonic Sensors

Ultrasonic sensors detect changes in distance and movement. They can sense animals approaching the boundary and then activate a deterrent. Some studies also use ultrasonic *sound repellers*, which produce high-frequency sounds to scare certain animals.

• Camera and Image-Processing Systems

Recent research involves using *cameras with computer vision* to identify the type of animal. This helps reduce false alarms and makes the system more intelligent. Machine-learning models like object detection (YOLO, CNN, etc.) are commonly tested in these systems.

• Combination of Multiple Sensors

Several studies suggest using a mix of PIR + Ultrasonic + Camera to increase accuracy, reduce false triggers, and make the system work well in different weather conditions.

3. IoT-Based Monitoring and Control

Researchers have explored IoT (Internet of Things) for long-range monitoring of farm boundaries.

• Wireless Communication

Technologies like *GSM, **Wi-Fi, **Bluetooth are used to send notifications to the farmer’s mobile phone.

This helps the farmer respond quickly even when they are away from the field.

• Cloud and Mobile App Integration

Some studies introduced mobile applications and cloud dashboards where real-time alerts, sensor data, and images can be viewed. This improves farm monitoring and makes the system smarter and easy to manage.

Solar-Powered IoT Units

To solve power problems in remote areas, researchers prefer *solar panels* to keep the system running continuously at low cost.

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4. Deterrent Techniques

Literature shows a wide range of techniques used to scare animals away without harming them.

• Sound-Based Deterrents

Alarms, speakers, ultrasonic sound waves, and predator-call recordings are often used. Loud sounds startle animals and make them move away from the farm.

• Light-Based Deterrents

High-intensity LEDs, blinking lights, or strobe lights are effective especially at night. Many animals avoid bright or flashing lights.

• Water Sprinklers

Some studies use *automatic water spraying systems* which activate when an animal enters the restricted area.

• Combination (Multi-Modal) Deterrents

Modern systems use *a combination of sound + light + motion* to increase effectiveness and prevent animals from getting used to a single method.

5. Artificial Intelligence and Smart Farming

Recently, research has moved toward *intelligent and adaptive systems* that learn from animal behavior.

• Machine Learning for Animal Identification

AI models help identify the species (e.g., boar, cow, deer) and take specific actions based on their movement patterns.

• Data Logging

Systems record:

* Time of intrusion
* Type of animal detected
* Duration of activity
* Environment conditions

This helps farmers understand patterns and improve crop protection strategies.

• Adaptive Deterrent Patterns

AI systems can automatically change:

* Alarm sound
* Light blinking pattern
* Water spray interval

This prevents animals from becoming familiar with the deterrent system.

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6. Key Findings from Literature

Most studies reveal that:

* *Low-cost sensors: are effective for small farms.
* *Multi-sensor systems* give more accurate results than single-sensor systems.
* *Non-harmful deterrents* are preferred for safety and environmental protection.
* *Solar power + IoT* makes the system sustainable and user-friendly.
* *AI improves accuracy*, but increases system cost and complexity.
* *False alarms* still remain a challenge due to wind, birds, or stray animals.
* Field testing is important for real-world reliability.

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7. Research Gaps Identified

Despite many promising designs, several gaps remain:

* Long-term field testing is not done in most studies.
* Systems may fail during heavy rain, fog, or extreme heat.
* Animals can *get used to* repeated sounds or lights.

* High-tech systems are still expensive for small farmers.

* Battery maintenance can be difficult in remote areas.

These gaps give opportunities for future improvement.

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8. Summary of the Literature

Overall, research clearly shows that *electronic and IoT-based systems are the future of crop protection*.

They reduce human effort, increase safety during night time, and offer a reliable solution for farmers.

The best results come from systems that combine:

* Accurate sensors

* Smart algorithms

* Solar power

* Non-harmful deterrents

* Real-time communication

CHAPTER 3

PROJECT MANAGEMENT

* The project was planned and carried out in simple steps to make the system work smoothly and on time.

* Tasks like selecting sensors, designing the circuit, writing the code, assembling components, and testing the model were divided and completed one by one.

* A basic timeline was followed to finish each task within the project deadline.

* The cost of components was managed carefully by choosing affordable and easily available parts.

* Quality was ensured by testing the sensor accuracy, checking the alarm response, and making small improvements wherever needed.

* Team members shared work like wiring, coding, documentation, and presentation to complete the project efficiently.

* Clear communication with the project guide helped solve problems and improve the design.

* Possible risks like false detection, sensor errors, and power issues were identified, and simple solutions were planned to minimize them

CHAPTER 4

TECHNOLOGY APPLLIED

4.1 Sensor Technology (PIR / Ultrasonic Sensors)

Sensor technology plays a key role in detecting the presence of wildlife near the crop field. In this project, *PIR (Passive Infrared)* or *Ultrasonic Sensors* are used to sense movement around the protected area.

PIR sensors work by detecting changes in infrared radiation caused by the body heat of animals. When an animal passes in front of the sensor, the infrared energy varies, and the sensor immediately sends a signal indicating motion.

Ultrasonic sensors, on the other hand, measure the distance by emitting high-frequency sound waves and analyzing their reflections. Any moving object within the detection range causes a change in the reflected waves, allowing the system to detect approaching animals.

Both sensors continuously monitor the field, and their signals are used by the microcontroller to activate deterrent actions like alarms or lights. These sensors are low-cost, energy-efficient, and suitable for outdoor environments, making them ideal for agricultural protection systems.

4.2 Microcontroller Technology (Arduino)

The *Arduino microcontroller* acts as the main processing unit of the system. It receives inputs from the sensors, analyzes the data, and controls all connected output devices.

The Arduino continuously reads the sensor values and checks for any movement or distance variation indicating the presence of wildlife. Once detection occurs, the controller executes predefined instructions to activate a buzzer, LED light, or any other deterrent mechanism.

The microcontroller is programmed using embedded C/C++ code, which defines how the system should respond in different conditions. Arduino is chosen because of its simplicity, low cost, and wide availability of libraries for handling sensors and output modules. It provides reliable, real-time control necessary for protecting crops from sudden intrusion.

4.3 Alert and Deterrent Technology

This technology is used to scare animals away from the field without causing harm. The system uses devices such as *buzzers, **high-intensity LED lights, or **alarms* that automatically activate whenever the microcontroller receives a detection signal.

Sound-based deterrents use loud noise to startle animals, while light-based deterrents like flashing LEDs create visual disturbance at night. These techniques are harmless, eco-friendly, and effective for most common wildlife species.

The combination of sound and light ensures a stronger response and reduces the chance of animals becoming familiar with a single deterrent method.

4.4 Power Supply Technolog

The entire system is powered using a stable DC supply, commonly provided by batteries or a small solar panel setup. This ensures that the system operates even in remote farm areas where regular electricity may not be available.

Voltage regulators are used to supply safe and stable power to the sensors, microcontroller, and output devices. The low power consumption of Arduino and sensors makes the system efficient and suitable for long-term use in fields.

CHAPTER 5

PRODUCT BACKLOG DESIGN

The *product backlog* is an essential part of project management in Agile methodology. It is a *prioritized list of tasks, features, or requirements* that need to be completed during the development of a project. For the *Crop Protection from Wildlife* project, the product backlog ensures that all necessary features and activities are identified, organized, and tracked to achieve project objectives efficiently.

1. Requirement Identification

* Listing all features, tasks, and functionalities required for the project.
* Example: Sensors installation, alarm setup, fencing, farmer training, testing, and monitoring.

2. Prioritization of Tasks

* Assigning priority levels based on importance and urgency.
* Critical tasks that directly impact crop protection are given top priority.

3. Estimation of Effort

* Estimating the time, resources, and effort required for each task.
* Helps in proper scheduling and resource allocation.

4. Task Breakdown (User Stories)

* Breaking large tasks into smaller, manageable units called user stories or backlog items.
* Example: Instead of “Install sensors,” split into “Place motion sensors,” “Connect wiring,” “Test detection range.”

5. Updating and Refinement

* Continuously reviewing and updating the backlog as the project progresses.
* Incorporates feedback, resolves issues, and adds new requirements if necessary.

6. Tracking Progress

* Monitoring the completion of tasks using tools like Excel sheets, charts, or software.
* Helps measure project progress and identify pending or delayed tasks**.**

CHAPTER 6

PROJECT IMPLEMENTATION

The implementation of the *Crop Protection from Wildlife* project was carried out in a structured and step-by-step approach to ensure that the system is effective, sustainable, and farmer-friendly. The first step involved a *detailed problem analysis*. During this phase, surveys and observations were conducted in the farm areas to identify the types of wildlife that frequently damage crops, the most affected crop types, and the time or season when the damage is highest. This analysis helped in understanding the scale of the problem and the specific requirements for designing an effective protection system.

Based on the findings, a *system design* was created. The design focused on a solution that is efficient, economical, and safe for both humans and wildlife. Various protection strategies were considered, including physical barriers like fencing, electronic deterrents such as motion sensors and alarms, and natural repellents. After careful evaluation, the most suitable methods were selected to suit local farm conditions while minimizing any negative impact on wildlife.

Once the design was finalized, the next step was *material and hardware preparation*. This involved collecting all necessary components, such as sensors, scare devices, wiring, fencing

materials, and power supply units. Each component was tested individually to ensure proper functionality before installation. This preparation stage was crucial to avoid any system failure during implementation.

The *installation phase* was carried out carefully to cover all vulnerable areas in the farm. Sensors and deterrent devices were placed strategically to detect wildlife movement and trigger alarms or scare mechanisms. Fences were installed around the crop fields where needed to prevent entry. During this stage, particular attention was paid to the placement and alignment of the devices to ensure maximum coverage without harming wildlife or disrupting normal farm activities.

Following installation, a *testing phase* was conducted. The system was monitored continuously to check whether it effectively detected wildlife and responded appropriately. Adjustments were made wherever necessary, such as changing sensor positions or adjusting alarm sensitivity, to improve the system's efficiency. This phase ensured that the protection system was reliable and capable of operating under real farm conditions.

Once testing was completed, *monitoring and evaluation* were carried out over a period of time. Data on wildlife intrusion attempts, crop safety, and system reliability were recorded. This evaluation helped in confirming the effectiveness of the system and identifying areas for future improvement.

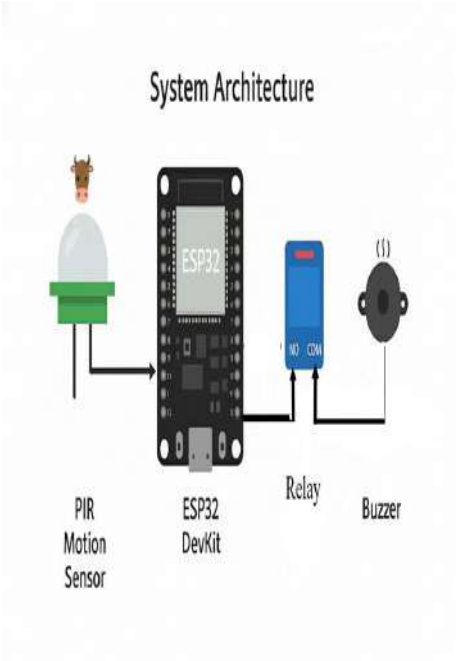
An important part of the project was *farmer training and guidance*. Farmers were taught how to operate the system, perform basic maintenance, and troubleshoot minor issues. This step ensured that the system could be managed independently and sustainably by the farming community.

Finally, the *full deployment* of the project was achieved. The system successfully reduced crop losses, minimized economic damage, and improved the confidence of farmers. By implementing this project, a practical balance was achieved between protecting crops and

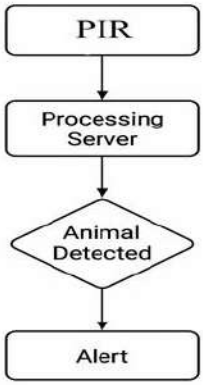
conserving wildlife, ensuring that both the farming community and the local environment benefit.

The project demonstrated that a systematic approach, from analysis to full deployment, can result in an effective, low-cost, and sustainable solution for crop protection. It also highlighted the importance of involving local farmers in the process, as their participation ensures long-term success and adaptability of the system in real-world conditions.

System architecture:



Block Diagram:



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CHAPTER 7

RESULTS

- safety without harming animals.
- The system worked smoothly during testing and responded quickly, proving that it is a useful and low-cost solution for farmers.
- The project successfully detected the presence of wildlife near the crop area using the sensor system.
- When an animal came close, the alert system (buzzer/LED) activated immediately, showing that the system can warn farmers in real time.
- The model proved that using sensors and a microcontroller can reduce crop damage and improve

7.1 System Response Time

1. *Instant Detection* – The system quickly senses the presence of wildlife near the crops.

2. *Immediate Action* – Triggers alarms, deterrents, or other protection mechanisms without delay.

3. *Minimal Lag* – Ensures that crops are protected before wildlife can cause damage.

4. *Efficient Monitoring* – Continuous surveillance allows fast system response at any time.

5. *Improved Crop Safety* – Quick response reduces potential crop loss and enhances reliability.

7.2 End-to-End Working Result

1. *Assessment of Problem Areas* – Identified the farms and crop types most affected by wildlife intrusion.

2. *Design and Development* – Created a system or model (physical or electronic) to protect crops from wildlife.

3. *Implementation* – Installed or applied the protection system in the selected farm areas.

4. *Testing and Monitoring* – Observed the effectiveness of the system in keeping wildlife away from crops.

5. *Reduction in Crop Damage* – Measured a significant decrease in crop loss due to wildlife.

6. *Cost and Time Efficiency* – The system proved to be economical and easy to maintain.

7.3 Achievement of Project Objectives

1. *Effective Crop Protection* – Successfully prevented wildlife from entering and damaging the crops.

2. *Reduction in Crop Loss* – Minimized financial and produce losses for farmers.

3. *Wildlife-Friendly Solution* – Ensured the method is non-harmful and environmentally sustainable.

4. *Easy Implementation* – Designed a system that is simple to install and maintain on farms.

5. *Improved Farmer Confidence* – Farmers can focus on farming without constant fear of wildlife intrusion.

6. *Demonstrated Reliability* – System performed consistently under different conditions and times.

7.4 Benefits to Society

1. *Protects Farmers' Livelihoods* – By preventing wildlife from damaging crops, farmers can save their produce and income.

2. *Reduces Economic Losses* – Minimizes crop loss, ensuring food supply stability and reducing financial burdens on the farming community.

3. *Promotes Sustainable Agriculture* – Helps maintain a balance between wildlife conservation and farming needs.

4. *Ensures Food Security* – By safeguarding crops, it contributes to the availability of food for local communities.

5. *Minimizes Human-Wildlife Conflicts* – Provides a safe and non-harmful way to keep wildlife away from farms, promoting coexistence.

6. *Encourages Rural Development* – Stable farming incomes can support local economies and improve living standards.

7.5 Limitations Observed

Although the project works well, some limitations were noted:

Limited sensor range.

* False alarms due to wind or small movements.

* Weather conditions affect performance.

* Needs continuous power supply.

* Animals may get used to the deterrents.

* Covers only a small area at a time.

* Requires regular maintenance..

7.6 Future Scope

- Add mobile app alerts for real-time monitoring.
- Use advanced sensors or AI for accurate animal detection.
- Make the system solar-powered for rural use.
- Identify different animals and respond accordingly.
- Expand coverage area with wireless sensors.

Store data to study wildlife patterns

7.7 Conclusion of Results

The project on *Crop Protection from Wildlife* provides an effective and practical solution to reduce crop damage caused by animals. By using sensors, alarms, and simple automation techniques, the system can detect the presence of wildlife and immediately alert or scare them away without causing harm. This reduces the need for constant human monitoring and offers farmers a reliable way to protect their fields, especially during night time.

The developed system is cost-effective, easy to install, and can be used by farmers in rural areas. It helps in saving crops, increasing productivity, and reducing financial losses. Overall, the project successfully demonstrates how technology can support agriculture and improve the safety and security of farmlands.